

JNTU Kakinada

SUPPLEMENTARY EXAMINATION ? 2025

Theory of Machines ? Supplementary Examination, 2025

Subject Code: ME401 Semester: 4 Regulation: R23

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Mechanisms and Machines)

1. Describe the working principle of Mechanisms and Machines with necessary equations. [12 Marks]
2. Compare and contrast the key concepts involved in Mechanisms and Machines. [12 Marks]
3. How is Mechanisms and Machines implemented? Give an algorithm / procedure and explain. [8 Marks]

SECTION B (Unit 2: Velocity and Acceleration Analysis)

4. Solve a problem related to Velocity and Acceleration Analysis. Show all the steps clearly. [12 Marks]
5. Describe the working principle of Velocity and Acceleration Analysis with necessary equations. [12 Marks]
6. Derive the expression related to Velocity and Acceleration Analysis and explain each term. [8 Marks]

SECTION C (Unit 3: Cams and Followers)

7. Write short notes on Cams and Followers. [12 Marks]
8. Compare and contrast the key concepts involved in Cams and Followers. [8 Marks]
9. Define Cams and Followers. Explain its significance in Theory of Machines with a suitable example. [12 Marks]

SECTION D (Unit 4: Gears and Gear Trains)

10. Compare and contrast the key concepts involved in Gears and Gear Trains. [12 Marks]
11. Solve a problem related to Gears and Gear Trains. Show all the steps clearly. [12 Marks]
12. Explain Gears and Gear Trains with a neat diagram/sketch and a suitable example. [12 Marks]

SECTION E (Unit 5: Balancing and Vibration)

13. How is Balancing and Vibration implemented? Give an algorithm / procedure and explain. [8 Marks]

14. Derive the expression related to Balancing and Vibration and explain each term.

[12 Marks]

15. Discuss the engineering applications of Balancing and Vibration. [12 Marks]

*** End of Question Paper ***